UNDERSTANDING HUMAN VISUAL ATTENTION THROUGH GAZE TRAJECTORIES IN NATURAL IMAGES

ABSTRACT

This study explores how humans distribute their visual attention while freely viewing natural photographs. Using the COCO-FreeView eye-tracking dataset, the study analyses gaze fixation coordinates and recorded fixation durations from 10 participants. The analysis focuses on first post-centre fixations, overall spatial fixation patterns, similarities in gaze distributions across participants, and fixation duration characteristics. Python libraries, including Pandas, NumPy, and Matplotlib, were used for data processing, statistical analysis, and visualisation. The results indicate a concentration of initial fixations near the centre of the images, while subsequent fixation patterns show a broader spatial distribution. Participant comparisons reveal both shared spatial patterns and individual differences in visual attention. The study provides an exploratory analysis of human visual behaviour when viewing natural images without a specific visual target.

Keywords: Visual Attention, Eye Tracking, Gaze Trajectories, Fixation Analysis, COCO-FreeView, Human Visual Behaviour.

1. INTRODUCTION

Human vision is one of the primary ways through which people interact with and understand their surroundings. Every day, individuals observe a wide variety of visual information, including natural landscapes, objects, people, animals, buildings, and other elements of the environment. However, the human visual system does not process every part of a scene with equal attention. Instead, visual attention is distributed selectively, allowing individuals to focus on particular regions while exploring the rest of the scene.

When a person views a photograph, their eyes move across different locations within the image. Some regions may attract attention quickly, while others may receive attention later or remain unexplored. Understanding these patterns provides valuable insights into how humans visually explore and interpret complex scenes.

1.2 Eye Tracking and Visual Attention

Eye tracking is a research technique used to record and analyse eye movements. It provides information about where a person looks, how their gaze moves, and how long their eyes remain relatively stable at particular locations.

Two important concepts in eye movement analysis are fixations and saccades. A fixation is a period during which the eyes remain relatively stable while visual information is processed. A saccade is a rapid eye movement that shifts the gaze from one location to another.

The sequence of fixations and saccades forms a gaze trajectory, commonly referred to as a scan path.

By analysing these eye movement patterns, researchers can investigate how people distribute their visual attention across images. Fixation locations help identify areas receiving visual attention, while fixation durations provide information about how long viewers remain focused on particular locations.

1.3 Visual Attention in Natural Images

Natural photographs contain multiple visual elements that can attract human attention. For example, an image may contain people, animals, trees, buildings, roads, or landscapes. Different viewers may direct their attention towards different regions depending on the visual information present in the scene.

Although some viewing patterns may be shared across individuals, visual exploration can also vary between participants. Studying these similarities and differences can help researchers understand whether people tend to focus on similar spatial regions or develop individual viewing patterns.

Free-viewing experiments are particularly useful for investigating natural visual behaviour because participants are allowed to observe images without being instructed to search for a specific object or location. This provides an opportunity to study how attention is naturally distributed.

1.4 Motivation for the Study

Understanding human visual attention is relevant to several research areas, including computer vision, cognitive science, human-computer interaction, and artificial intelligence. Eye-tracking data can help researchers investigate how humans interact with visual information and provide useful reference points for studying computational attention models.

Analysing gaze trajectories also presents an opportunity to explore the relationship between shared visual attention and individual differences. Rather than focusing only on the objects present in an image, this study examines the spatial distribution and temporal characteristics of recorded eye fixations.

1.5 Overview of the Present Study

This project presents an exploratory analysis of human visual attention using the COCO-FreeView eye-tracking dataset. The dataset contains gaze fixation observations collected from participants while they freely viewed natural photographs.

The analysis focuses on four main aspects: the location of the first post-centre fixation, the overall spatial distribution of fixations, similarities in gaze distributions between participants, and fixation duration patterns.

Python-based data analysis techniques are used to process the fixation coordinates, calculate descriptive statistics, compare participant-level gaze distributions, and generate visualisations such as fixation heatmaps and similarity matrices.

Through these analyses, the study aims to identify general patterns in human visual exploration while also examining variations between individual participants.

2. RESEARCH OBJECTIVES

The primary objective of this study is to explore how humans distribute their visual attention while freely viewing natural photographs using eye-tracking data.

The specific objectives are:

1. **First Fixation Analysis:** To examine where participants direct their gaze immediately after the initial centre fixation.
2. **Spatial Attention Analysis:** To identify regions of images that receive higher concentrations of visual fixations.
3. **Participant Comparison:** To investigate similarities and differences in gaze distributions across multiple participants.
4. **Fixation Duration Analysis:** To examine the duration of recorded fixations and understand the distribution of viewing times.
5. **Visualisation of Gaze Patterns:** To represent fixation distributions and participant similarities through heatmaps, statistical summaries, and other visualisations.

Overall, the study aims to develop an understanding of shared and individual patterns of human visual attention during free viewing.

3. RESEARCH QUESTIONS

This study is guided by the following research questions:

1. **RQ1:** Where do participants direct their first gaze after the initial centre fixation when viewing natural photographs?
2. **RQ2:** Which spatial regions receive the highest concentration of visual fixations across the observed images?
3. **RQ3:** To what extent do participants show similar or different gaze distribution patterns when viewing the same photographs?
4. **RQ4:** How are fixation durations distributed during free viewing, and what patterns can be observed in the recorded viewing times?

These questions guide the exploratory analysis of first fixations, overall spatial attention, participant-level similarities, and fixation duration patterns.

4. DATASET DESCRIPTION

4.1 Dataset Overview

This study uses the COCO-FreeView eye-tracking dataset, which contains eye movement observations collected while participants freely viewed natural photographs. The dataset is designed to support research into human visual attention when no specific object or viewing target is provided.

According to the dataset documentation, participants viewed images for a fixed duration of five seconds. Their gaze positions were recorded as fixation coordinates, providing information about the spatial distribution of visual attention.

4.2 Dataset Characteristics

The dataset documentation describes the following characteristics:

- Dataset name: COCO-FreeView
- Participants: 10 university students
- Viewing condition: Free viewing
- Image resolution: 1680 × 1050 pixels
- Viewing duration: 5 seconds per image
- Recorded information: Fixation coordinates and associated timing information

4.3 Data Used in This Study

For this project, the downloaded training and validation JSON files were used for analysis.

- Training records: 30,129
- Validation records: 6,457
- Total records: 36,586
- Participants represented: 10
- Common images viewed by all participants: 3,524

The dataset records contain information such as image names, participant identifiers, fixation coordinates (X and Y), timing values (T), and fixation sequence length.

The analysis was performed using the available training and validation records. The test split was not included in this study.

4.4 Dataset Source and Download Information

The COCO-FreeView dataset was originally documented through the COCO-Search18 research project. The original dataset download endpoint was unavailable during the data collection stage of this project.

Therefore, the training and validation JSON files were obtained from a third-party archive hosted by the Artificial Intelligence and Bioinformatics Laboratory (AILB), University of Modena and Reggio Emilia, Italy.

Archive URL: https://ailb-web.unimore.it/publicfiles/ScanDiff_ICCV2025/coco_freeview.zip

The downloaded files used in this study were:

- COCOFreeView_fixations_train.json
- COCOFreeView_fixations_validation.json

Only these two files were used for the analysis. The test split was not downloaded.

The downloaded archive contained a JSON structure that differed slightly from the fields described in the original dataset documentation. Therefore, the analysis was performed using the fields available in the downloaded records, and this difference is acknowledged as a dataset provenance limitation.

5. TOOLS AND TECHNOLOGIES

This study uses Python-based tools and libraries for data processing, statistical analysis, and visualisation of eye-tracking data.

The following tools and technologies were used:

1. Python

Python was used as the primary programming language for loading, processing, analysing, and interpreting the dataset.

2. Google Colab

Google Colab provided a cloud-based notebook environment for executing Python code and conducting the analysis.

3. Pandas

Pandas was used to load JSON records, organise fixation observations into DataFrames, perform data manipulation, and calculate descriptive statistics.

4. NumPy

NumPy was used for numerical computations, coordinate calculations, spatial distance measurements, and array operations.

5. Matplotlib

Matplotlib was used to generate visual representations of the analysis, including fixation heatmaps, duration distributions, and participant comparison plots.

6. JSON

JSON files were used as the input data format for storing and accessing eye-tracking records.

Together, these tools provided the computational environment required to conduct the exploratory analysis and generate the research results.

6. DATA PREPARATION AND PREPROCESSING

6.1 Data Loading

The training and validation datasets were loaded from their respective JSON files using Python. Both datasets were combined to create a single collection of records for the exploratory analysis.

6.2 Data Inspection

The loaded records were examined to understand their structure and identify the available variables. The analysis used fields such as image name, participant identifier, fixation coordinates, timing values, and fixation sequence length.

The combined dataset contained 36,586 records, consisting of 30,129 training records and 6,457 validation records.

6.3 Coordinate Verification

The fixation coordinates were examined in relation to the image dimensions of 1680 × 1050 pixels. The initial coordinate in each record was also checked to evaluate whether it was located near the image centre. The mean initial coordinates were approximately (838.31, 520.13), while the image centre was (840, 525). The mean distance from the image centre was approximately 31.27 pixels.

This supported the assumption that the first coordinate represented the initial centre fixation.

6.4 First Post-Centre Fixation Extraction

According to the dataset documentation, the initial centre fixation is included in the recorded coordinate sequences. Therefore, the coordinate at index 1 was used to identify the first post-centre fixation.

Records with insufficient coordinate information were excluded from this particular extraction. A total of 36,544 first post-centre fixations were extracted from the 36,586 records.

6.5 Overall Fixation Extraction

For the overall fixation analysis, all recorded coordinates after the initial centre fixation were extracted from each trial. The corresponding timing values were retained for duration analysis.

This produced a collection of 528,476 post-centre fixation observations.

6.6 Participant and Image Matching

Participant identifiers and image names were used to organise observations for participant-level comparisons. The intersection of images viewed by all ten participants contained 3,524 common images. These common images were used to compare spatial gaze distributions across participants.

7. METHODOLOGY AND ANALYSIS TECHNIQUES

The study follows an exploratory data analysis approach to investigate human visual attention using eye-tracking observations. The analysis was conducted using Python and its scientific computing libraries.

7.1 First Fixation Analysis

The first post-centre fixation from each trial was extracted to examine where participants initially directed their attention after the initial centre fixation.

The X and Y coordinates were used to calculate the spatial distribution of these first fixations. A two-dimensional heatmap was generated to visualise regions receiving higher concentrations of first fixations.

7.2 Overall Spatial Fixation Analysis

All post-centre fixation coordinates were extracted from the training and validation records.

A two-dimensional histogram-based heatmap was generated using the X and Y coordinates. This visualisation helped identify spatial regions with higher concentrations of recorded fixations across the analysed observations.

7.3 Participant-Level Analysis

Fixation observations were grouped according to participant identifiers to examine individual gaze distributions.

Descriptive statistics, including mean fixation coordinates and distance from the image centre, were calculated for each participant. These measurements helped identify similarities and variations in spatial viewing patterns.

7.4 Gaze Distribution Similarity Analysis

To compare participants viewing the same photographs, spatial fixation maps were generated for each participant-image combination.

Each spatial map was divided into a grid of 12 columns and 8 rows. The resulting spatial distributions were compared using cosine similarity.

The similarity scores were averaged across common images to construct a participant-level similarity matrix. A heatmap was then generated to visualise the similarities between participant pairs.

7.5 Fixation Duration Analysis

The recorded timing values associated with post-centre fixations were examined to understand the distribution of fixation durations.

Descriptive statistics, including the mean, median, minimum, maximum, and percentile values, were calculated. A histogram was generated to visualise the distribution of recorded fixation durations.

7.6 Statistical and Visualisation Methods

The study primarily uses descriptive statistical methods and graphical visualisations. These include coordinate averages, spatial distance calculations, two-dimensional histograms, cosine similarity, and fixation duration distributions.

The results are interpreted as exploratory observations rather than evidence of causal relationships or statistically significant differences.

8. RESULTS AND ANALYSIS

8. Results and Analysis

This section presents the findings obtained from the exploratory analysis of the COCO-FreeView eye-tracking dataset. The results are organised according to the research objectives and include numerical summaries and graphical representations of fixation patterns.

8.1 First Fixation Analysis

The first post-centre fixation was analysed to investigate where participants initially directed their visual attention after the initial centre fixation.

Out of 36,586 total records, 36,544 first post-centre fixations were successfully extracted. The remaining 42 records did not contain sufficient coordinate information for this analysis.

The overall mean first-fixation location was approximately X = 804.86 pixels and Y = 508.37 pixels. The mean distance from the image centre (840, 525) was approximately 227.43 pixels, while the median distance was 214.13 pixels.

The first-fixation heatmap showed a noticeable concentration of gaze locations around the central region of the images, particularly around X = 800–850 and Y = 480–520.

These observations suggest that participants frequently directed their initial post-centre gaze towards regions relatively close to the centre, although individual viewing patterns varied.

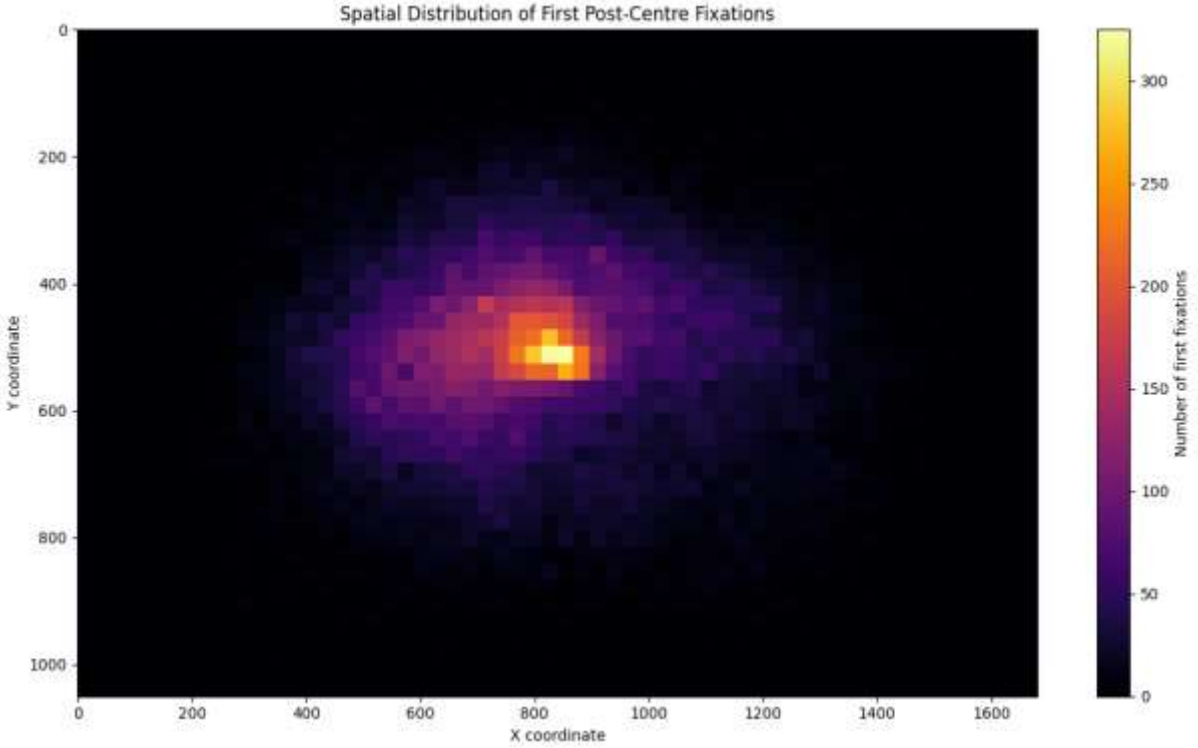


Figure 1: Heatmap showing the spatial distribution of first post-centre fixations across the analysed trials.

8.2 Overall Spatial Fixation Analysis

The overall fixation analysis examined 528,476 post-centre fixation observations collected across the training and validation records.

The spatial distribution of these fixations was visualised using a two-dimensional heatmap. The heatmap showed a strong concentration of fixations around the central region of the image, particularly near X = 800–850 and Y = 450–550. The distribution also showed a broader horizontal spread across the image.

The overall mean fixation coordinates were approximately X = 856.14 pixels and Y = 520.19 pixels. The mean distance from the image centre was 317.50 pixels, while the median distance was 300.68 pixels.

Compared with the first post-centre fixation analysis, the overall fixation observations showed greater average spatial dispersion. This indicates that gaze locations across the complete viewing sequences covered a wider range of spatial positions than the initial post-centre fixations.

These findings describe the spatial distribution of recorded fixations across different photographs and should not be interpreted as identifying specific objects or semantic regions.

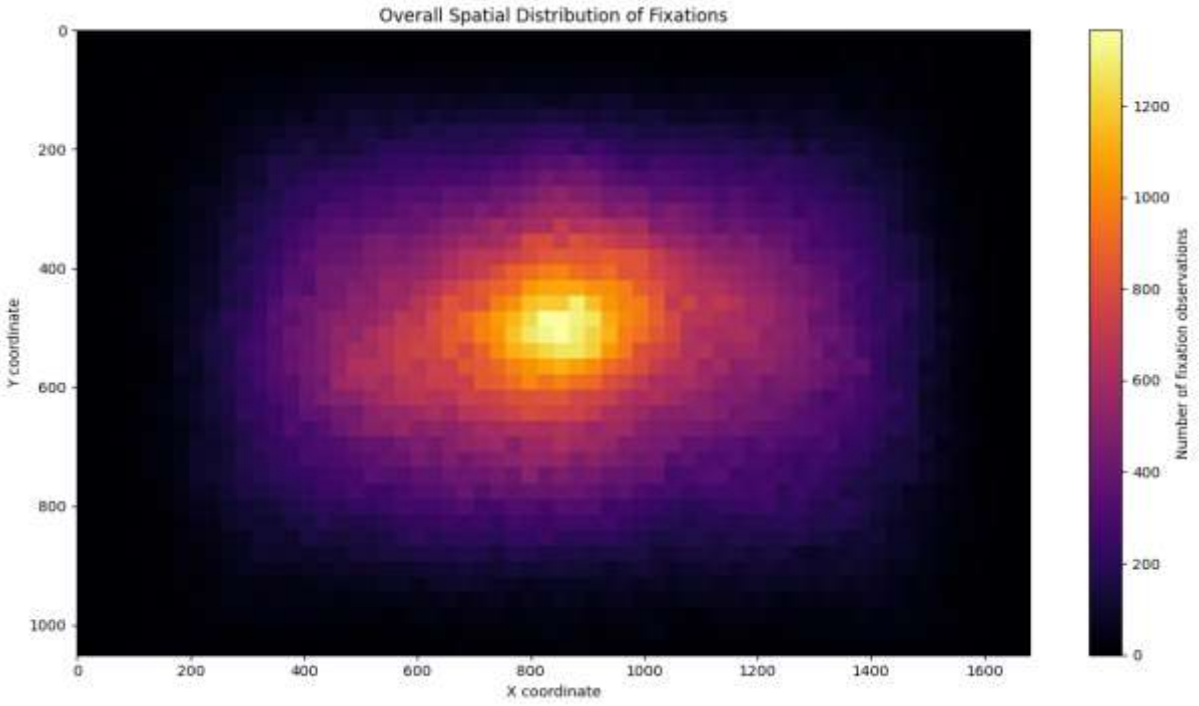


Figure 2: Heatmap showing the overall spatial distribution of post-centre fixations across all analysed trials.

8.3 Participant-Level Analysis

The participant-level analysis was conducted to examine individual differences in visual attention across the ten participants included in the dataset.

For each participant, the mean fixation coordinates and average distance from the image centre were calculated using all extracted post-centre fixation observations.

The results showed variations in the spatial distribution of gaze across participants. Participant 102 had the lowest mean distance from the image centre at approximately 257.51 pixels, whereas participant 109 had the highest mean distance at approximately 364.61 pixels.

These differences indicate that participants did not distribute their visual attention in exactly the same way. Some participants showed fixation patterns more concentrated around the image centre, while others exhibited greater spatial dispersion.

Participant-level comparisons provide insight into individual variations in visual exploration during free viewing.

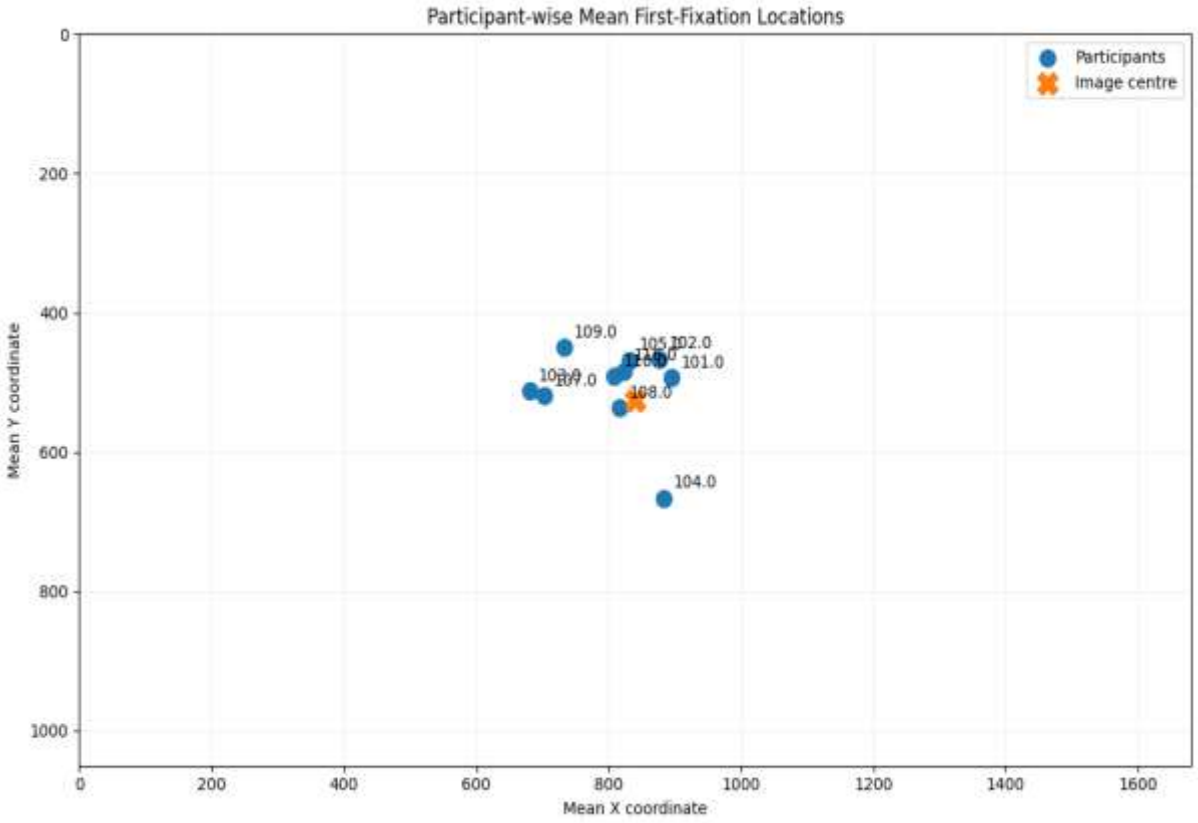


Figure 3: Comparison of mean fixation distance from the image centre across the ten participants.

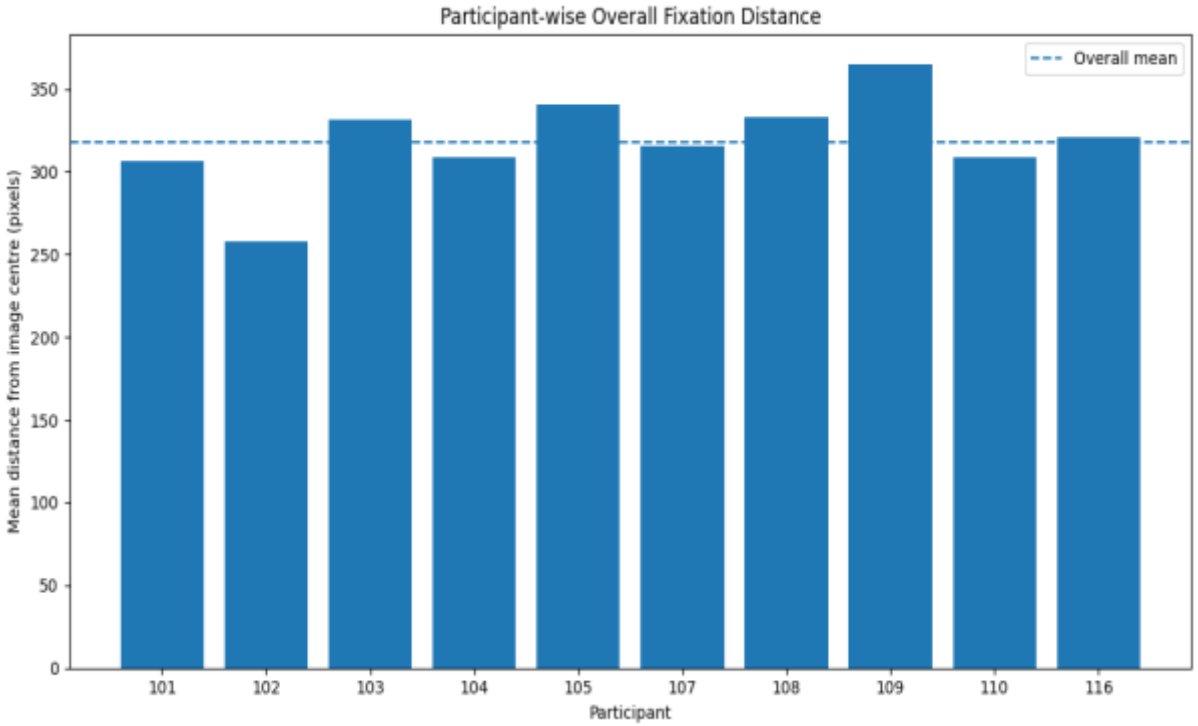


Figure 3. Comparison of mean fixation distance from the image centre across the ten participants.

8.4 Gaze Distribution Similarity Analysis

The gaze distribution similarity analysis was performed to investigate how similarly different participants viewed the same natural photographs.

A total of 3,524 images viewed by all ten participants were selected for comparison. For each participant-image combination, fixation coordinates were converted into a spatial representation using a grid consisting of 12 columns and 8 rows.

Cosine similarity was calculated between the spatial fixation maps of each participant pair. The resulting similarity scores were averaged across the common images to create a participant-level similarity matrix.

The mean pairwise similarity across the 45 unique participant pairs was approximately 0.3749, with a median similarity of 0.376.

The highest average similarity was observed between participants 103 and 107, with a score of approximately 0.4638. The lowest was observed between participants 102 and 104, with a score of approximately 0.2587.

These findings indicate partial spatial overlap in gaze distributions, alongside noticeable individual differences in visual attention. The similarity scores represent spatial distribution similarity and should not be interpreted as percentages of agreement.

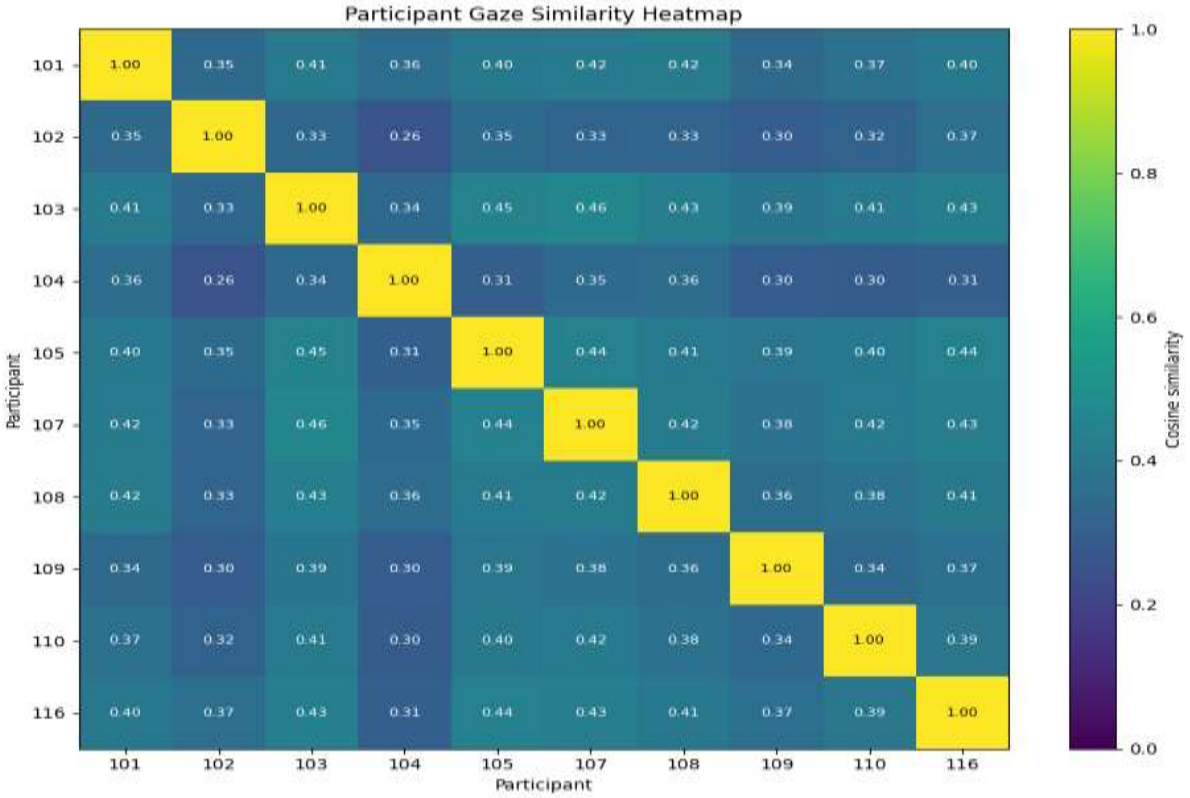


Figure 4. Heatmap showing pairwise gaze distribution similarity between the ten participants using cosine similarity.

8.5 Fixation Duration Analysis

The fixation duration analysis was conducted to examine how long participants maintained their gaze at recorded fixation locations during free viewing.

A total of 528,476 post-centre fixation duration observations were included in this analysis. The recorded durations ranged from 2 milliseconds to 5,006 milliseconds.

The mean fixation duration was approximately 262.04 milliseconds, while the median duration was 230 milliseconds. The 95th percentile was 525 milliseconds, indicating that most recorded durations were below this value.

The distribution showed a concentration of observations around the 150–300 millisecond range, with a right-skewed pattern caused by a smaller number of longer recorded durations.

These findings describe the distribution of recorded fixation durations and provide an overview of temporal characteristics in the participants' visual exploration patterns.

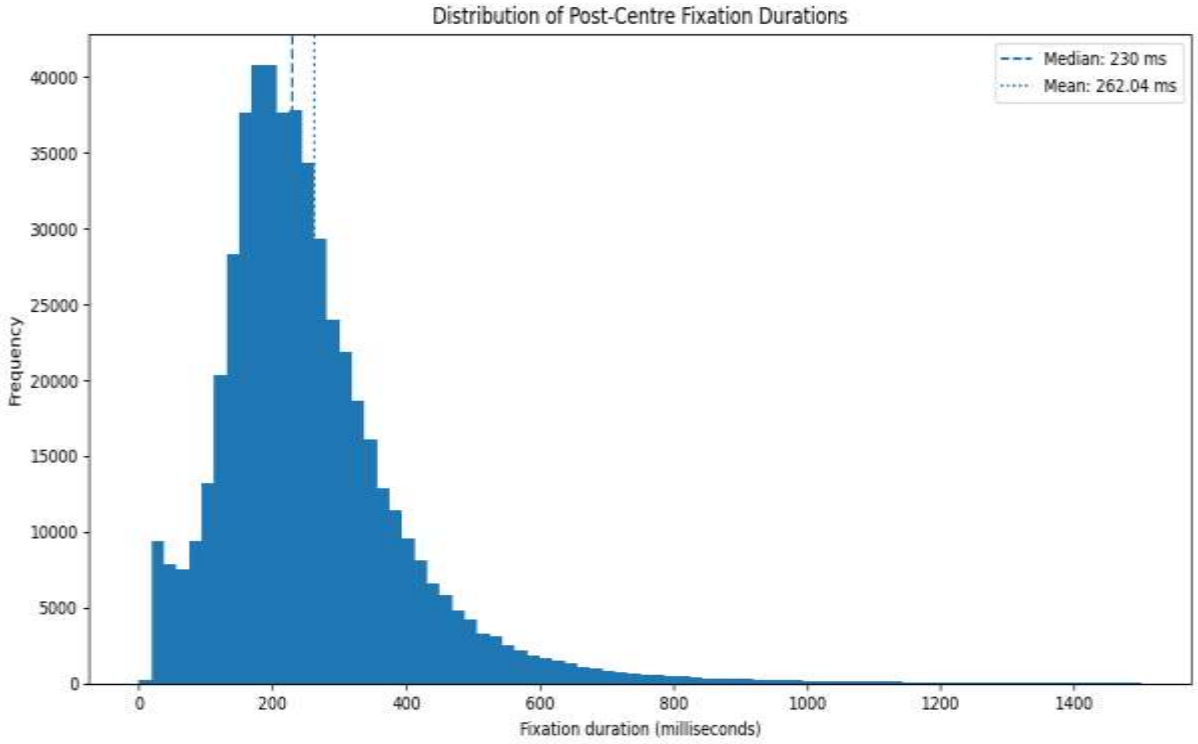


Figure 5. Histogram showing the distribution of fixation durations across all analysed post-centre fixations.

9. DISCUSSION

The findings of this study provide an exploratory understanding of how people distribute their visual attention while freely viewing natural photographs.

The first-fixation analysis showed that participants frequently directed their initial post-centre gaze towards regions relatively close to the centre of the images. However, the overall fixation analysis revealed a broader spatial distribution, suggesting that visual exploration extended beyond the initial viewing locations.

The participant-level analysis highlighted individual differences in gaze behaviour. Variations in mean fixation distances indicate that participants differed in how their fixations were spatially distributed across the images.

The gaze similarity analysis further demonstrated that participants shared some spatial patterns while also exhibiting individual variations. The average cosine similarity of approximately 0.3749 suggests partial overlap between participant-level spatial fixation distributions. However, these values do not represent exact agreement between viewers.

The fixation duration analysis showed that recorded fixation durations were concentrated around shorter time intervals, with a smaller number of longer observations. This provides additional information about the temporal characteristics of visual attention during free viewing.

Overall, the results suggest that human visual attention involves both shared spatial tendencies and individual differences. The study demonstrates how eye-tracking data and exploratory statistical methods can be used to investigate these patterns across natural photographs.

10. LIMITATION OF STUDY

Although this study provides useful observations about human visual attention, several limitations should be considered when interpreting the findings.

1. Dataset Coverage

The analysis was conducted using the available training and validation records rather than the complete dataset. Therefore, the findings represent only the downloaded portions of the dataset.

2. Participant Sample

The dataset contains observations from ten participants. This relatively small participant group limits the extent to which the findings can be generalised to a wider population.

3. Spatial Analysis

The fixation heatmaps represent spatial distributions across image coordinates. They do not independently identify specific objects or semantic regions within photographs.

4. Similarity Measurement

Cosine similarity was calculated using spatial maps divided into a 12 × 8 grid. This provides a coarse comparison of gaze distributions and may not capture fine-grained differences in fixation locations.

5. Dataset Format and Provenance

The downloaded archive used in this analysis had a data schema that differed from the fields described in the original dataset documentation. This difference should be considered when interpreting the results, and the exact archive version and source should be documented for reproducibility.

6. Exploratory Nature of the Study

The analysis primarily uses descriptive statistics and visualisation techniques. It does not establish causal relationships or demonstrate statistically significant differences between participants.

Despite these limitations, the study provides an exploratory framework for examining spatial and temporal patterns of human visual attention using eye-tracking data.

11. CONCLUSION

This study explored human visual attention during free viewing of natural photographs using the COCO-FreeView eye-tracking dataset. The analysis focused on first post-centre fixations, overall spatial fixation distributions, participant-level gaze similarities, and fixation duration patterns.

The results showed that first post-centre fixations were concentrated relatively close to the centre of the images, while the overall fixation observations displayed a broader spatial distribution. Participant-level comparisons revealed variations in gaze locations, indicating individual differences in visual exploration.

The gaze similarity analysis identified partial spatial overlap between participants, suggesting that viewers may share certain visual attention patterns while also exhibiting individual differences. The fixation duration analysis provided additional insight into the temporal distribution of recorded gaze observations.

Through the use of Python-based data processing, statistical analysis, and visualisation techniques, this project demonstrates how eye-tracking datasets can be used to investigate human visual behaviour.

Overall, the study provides an exploratory understanding of spatial and temporal patterns in human visual attention. It also establishes a foundation for further investigation using larger participant groups, more detailed image-level analysis, and additional statistical methods.

REFERENCES

Conference, 56–61. https://doi.org/10.25080/Majora-92bf1922-00a Chen, Y., Yang, Z., Chakraborty, S., Mondal, S., Ahn, S., Samaras, D., Hoai, M., & Zelinsky, G. (2022). Characterizing target-absent human attention. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 5031–5040.

COCO-Search18. (n.d.). *COCO-FreeView*. https://sites.google.com/view/cocosearch/coco-freeview

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. *Nature, 585*, 357–362. https://doi.org/10.1038/s41586-020-2649-2

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering, 9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science*

Artificial Intelligence and Bioinformatics Laboratory, University of Modena and Reggio Emilia. (n.d.). *ScanDiff_ICCV2025 archive containing COCO-FreeView fixation files* [Dataset archive]. https://ailb-web.unimore.it/publicfiles/ScanDiff_ICCV2025/coco_freeview.zip